MASTER RUBRIC SYSTEM (MRS-02) — Python Programming

PART 1 — STRUCTURE

Each assignment uses a consistent rubric structure, but not every assignment must emphasize every category equally.

The goal is to support fast, repeatable grading while still aligning assessment to the actual arc of the course.

Performance Scale

Score	Level
4	**Exceeds**
3	**Meets**
2	**Developing**
1	**Needs Improvement**
0	Missing / Not Submitted

PART 2 — TECHNICAL COMPETENCIES

Technical competencies are course-specific and aligned to Python programming, debugging, data handling, AI-assisted development, and the capstone structure.

To keep grading manageable, this system uses **6 technical categories** rather than creating a separate category for every topic.

T1. Python Fundamentals

Evidence: syntax, variables, values, expressions, basic types, input/output

Level	Description
4 — Exceeds	Uses Python fundamentals accurately and confidently; code is clear, readable, and mostly free of basic syntax or type issues
3 — Meets	Uses Python fundamentals correctly with minor issues that do not prevent the program from working
2 — Developing	Uses some fundamentals correctly, but errors or misunderstandings affect reliability or clarity
1 — Needs Improvement	Struggles with basic syntax, values, variables, or input/output in ways that prevent meaningful completion

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T2. Program Logic and Control Flow

Evidence: conditionals, loops, boolean logic, sequencing, control flow

Level	Description
4 — Exceeds	Logic is clear, correct, and well-sequenced; conditionals and loops behave as intended across expected cases
3 — Meets	Logic mostly works with minor errors or limited edge-case handling
2 — Developing	Logic is partially correct but inconsistent, incomplete, or difficult to trace
1 — Needs Improvement	Program logic is missing, broken, or does not match the intended task

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T3. Code Organization and Structure

Evidence: functions, parameters, return values, collections, readability, structure choice, class-based recognition

Level	Description
4 — Exceeds	Code is well organized, readable, and appropriately structured; functions, collections, or simple class-based structures are used or interpreted effectively
3 — Meets	Code is generally organized and readable with minor structural issues
2 — Developing	Some organization is present, but code is repetitive, unclear, or inconsistently structured
1 — Needs Improvement	Code is disorganized, difficult to follow, or lacks meaningful structure

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T4. Data Handling and Integration

Evidence: lists, dictionaries, files, CSV/JSON, persistence, APIs, structured data, application-architecture recognition

Level	Description
4 — Exceeds	Handles data accurately and purposefully; uses appropriate structures, file/API patterns, or data representations with clear understanding
3 — Meets	Handles required data correctly with minor issues or limited refinement
2 — Developing	Data handling is partially successful but fragile, incomplete, or unclear
1 — Needs Improvement	Data is not stored, retrieved, parsed, or integrated correctly for the task

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T5. Debugging, Testing, and Validation

Evidence: error diagnosis, test cases, expected vs actual results, repair process, explanation of fixes

Level	Description
4 — Exceeds	Systematically identifies issues, tests behavior, validates fixes, and clearly explains why the solution works
3 — Meets	Finds and fixes most issues; uses reasonable checks or tests to confirm behavior
2 — Developing	Attempts debugging or testing but with limited success, incomplete validation, or unclear reasoning
1 — Needs Improvement	Does not effectively identify, test, or resolve problems

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T6. AI/RBA-Assisted Development and Accountability

Evidence: bounded AI use, RBA framing, prompt clarity, adaptation of AI output, AI-use justification, human decision-making

Level	Description
4 — Exceeds	Uses AI/RBA methods thoughtfully and transparently; clearly frames intent, evaluates output, adapts or rejects suggestions, and explains human decisions
3 — Meets	Uses AI appropriately when allowed and can explain the role AI played with minor gaps
2 — Developing	Uses AI with limited explanation, weak validation, or unclear distinction between AI contribution and student decision-making
1 — Needs Improvement	Relies on AI inappropriately, violates assignment limits, or cannot explain AI-assisted work

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PART 3 — CORE ABILITIES

Core abilities are behavioral and professional competencies that can be observed across assignments, labs, discussions, and presentations.

These categories can remain consistent across multiple courses.

C1. Solve Problems

Evidence: troubleshooting, persistence, strategy, independent recovery

Level	Description
4 — Exceeds	Consistently identifies, analyzes, and resolves problems independently or with well-targeted support
3 — Meets	Resolves most problems with reasonable effort and limited guidance
2 — Developing	Attempts problem solving but needs frequent support or direction
1 — Needs Improvement	Struggles to identify, analyze, or recover from problems

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C2. Communicate Clearly

Evidence: code explanation, project explanation, AI-use justification, presentation, discussion

Level	Description
4 — Exceeds	Clearly explains code, logic, decisions, and project outcomes; communicates reasoning with confidence and specificity
3 — Meets	Communicates adequately with minor gaps in clarity or detail
2 — Developing	Explanation is incomplete, vague, or difficult to follow
1 — Needs Improvement	Cannot clearly explain the work, logic, or decisions

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C3. Work Productively

Evidence: focus, task completion, use of lab time, organization, meeting deadlines

Level	Description
4 — Exceeds	Uses time effectively, stays organized, completes tasks reliably, and makes strong progress during lab time
3 — Meets	Generally productive with minor lapses in focus, organization, or completion
2 — Developing	Productivity is inconsistent or requires frequent redirection

Level	Description
1 — Needs Improvement	Frequently off-task, incomplete, or unable to make meaningful progress

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C4. Value Learning

Evidence: resource use, feedback application, curiosity, responsible AI/tool use, willingness to revise

Level	Description
4 — Exceeds	Actively uses resources, applies feedback, asks useful questions, and improves work through revision
3 — Meets	Uses resources and feedback appropriately with some guidance
2 — Developing	Uses resources inconsistently or applies feedback only partially
1 — Needs Improvement	Does not meaningfully engage with learning resources, feedback, or revision

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C5. Work Cooperatively

Evidence: peer support, pair/group activity, respectful collaboration, shared responsibility

Level	Description
4 — Exceeds	Actively contributes, supports peers, shares responsibility, and collaborates respectfully and effectively
3 — Meets	Participates appropriately with minor imbalance or limited initiative
2 — Developing	Collaboration is inconsistent, passive, or requires support
1 — Needs Improvement	Does not effectively collaborate or interferes with productive group work

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C6. Act Professionally

Evidence: attendance habits, engagement, respectful conduct, presentation quality, readiness

Level	Description
4 — Exceeds	Consistently professional, prepared, respectful, engaged, and accountable
3 — Meets	Generally professional with minor issues
2 — Developing	Professionalism, preparation, or engagement is inconsistent
1 — Needs Improvement	Behavior, preparation, or engagement does not meet course expectations

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PART 4 — SCORING MODEL

Schoology-Friendly Model

- Each category uses the same **0-4** scale
- Rubric rows can be reused across assignments
- Assignment emphasis can be adjusted by selecting the most relevant rows or weighting selected rows more heavily

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Total Categories

- 6 Technical Competencies
- 6 Core Abilities

If every category is used on an assignment:

48 points total = 12 categories × 4 points

However, most assignments should not need every row.

For grading efficiency, use the Assignment Mapping Matrix to select or emphasize the most relevant categories.

Recommended Practical Use

For smaller assignments:

- use 3-5 rubric rows
- emphasize the technical skill being practiced
- include 1-2 core ability rows when observable

For larger assignments and capstone work:

- use more rubric rows
- include AI/RBA accountability when relevant
- include communication and professionalism for presentations

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PART 5 — ASSIGNMENT MAPPING MATRIX (AMM-01)

How to Read This Matrix

Each assignment is mapped against:

- **Technical Competencies (T1-T6)**
- **Core Abilities (C1-C6)**

Legend

Symbol	Meaning
●●●	Primary focus / major grading emphasis
●●	Secondary focus
●	Light exposure / incidental
–	Not assessed or not relevant

Technical Competency Codes

Code	Description
T1	Python Fundamentals
T2	Program Logic and Control Flow
T3	Code Organization and Structure
T4	Data Handling and Integration
T5	Debugging, Testing, and Validation
T6	AI/RBA-Assisted Development and Accountability

Core Ability Codes

Code	Description
C1	Solve Problems
C2	Communicate Clearly
C3	Work Productively
C4	Value Learning
C5	Work Cooperatively
C6	Act Professionally

Assignment Mapping Table

Phase 1 — Foundations + Manual Habits

Assignment	T1	T2	T3	T4	T5	T6	C1	C2	C3	C4	C5	C6
A1 First Programs	●●●	●	–	–	●	–	●	●●	●●	●	●	●
A2 Decisions in Code	●●	●●●	–	–	●	–	●●	●	●●	●	●	●
A3 Loops and Repetition	●●	●●●	●	–	●●	–	●●	●	●●	●	●	●

Phase 2 — Structure + Code Literacy

Assignment	T1	T2	T3	T4	T5	T6	C1	C2	C3	C4	C5	C6
A4 Function Builder	●●	●●	●●●	–	●	●	●●	●	●●	●●	●	●
A5 List or Dictionary Mini-App	●●	●●	●●●	●●	●	●	●●	●	●●	●●	●	●
A6 Debug and Explain	●	●	●	–	●●●	●●	●●●	●●●	●	●●	●	●
A7 Reading Structured Code	●	●	●●●	●	●	●●	●●	●●	●	●●	●	●

Phase 3 — Data, Files, and Bounded AI Support

Assignment	T1	T2	T3	T4	T5	T6	C1	C2	C3	C4	C5	C6
A8 Save and Load Utility	●●	●	●●	●●●	●●	●●	●●	●	●●	●●	●	●
A9 Structured Data Reader	●	●	●	●●●	●●	●●	●●	●	●●	●●	●	●
A10 Data Representation and App-Structure Preview	●	–	●	●●●	●	●●	●	●●	●	●●	●	●
A11 API Data Fetcher	●	●●	●	●●●	●●	●●	●●●	●	●●	●●	●	●
A12 Python App Architecture Preview	●	–	●●	●●	●	●●	●	●●	●	●●	●	●

Phase 4 — RBA Mini-Unit + Capstone Application

Assignment	T1	T2	T3	T4	T5	T6	C1	C2	C3	C4	C5	C6
A13 RBA Project Framing Exercise	–	●	●●●	●●	●	●●●	●●	●●●	●●	●●●	●	●
A14 Capstone Proposal and Approval	●	●	●●●	●●	●	●●●	●●	●●●	●●	●●	●	●●
A15 Capstone Build	●●	●●	●●●	●●●	●●●	●●●	●●●	●●	●●	●●	●	●●

Assignment	T1	T2	T3	T4	T5	T6	C1	C2	C3	C4	C5	C6
A16 AI Use Justification and Final Presentation	●	●	●●	●	●●	●●●	●●	●●●	●●	●●	●	●●●

PART 6 — IMPLEMENTATION NOTES

In Schoology

Each row can become one rubric criterion.

Recommended workflow:

- create the full master rubric once
- duplicate or select relevant rows per assignment
- use the AMM to decide which rows should be emphasized
- keep the 0-4 scale consistent

Instructor Workflow

1. Identify the assignment's primary and secondary rubric rows from the AMM
2. Score the technical evidence first
3. Score core abilities only where observable
4. Add comments only when they clarify revision, misunderstanding, or exceptional performance

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Evaluation Frequency

Light scoring can occur weekly, but stronger evaluation should occur at natural checkpoints:

- Week 2: fundamentals and logic
- Week 4: debugging, testing, and code literacy
- Week 6: data/API/app-architecture recognition
- Week 7: RBA framing and proposal approval
- Week 8: capstone build, AI justification, and presentation

FINAL SYSTEM STATEMENT

This rubric system keeps grading consistent while allowing assignments to emphasize different parts of the course arc.

Students are assessed not only on whether their code runs, but also on whether they can reason through logic, organize code, handle data, debug and validate behavior, use AI responsibly, and explain their decisions clearly.